



Homography Decomposition Revisited

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Abstract

Homography refers to a specific type of transformation that relates two images of the same planar surface taken from different perspectives. Recovering motion parameters from a homography matrix is a classic problem in computer vision. It is important to derive a fast and stable solution to homography decomposition, since it forms a critical component of many vision systems, *e.g.*, in Structure-from-Motion and visual localization. The current state-of-the-art solvers can be categorized into two types of methods, the numerical procedures based on singular value decomposition (SVD), and the closed-form solution. The SVD-based methods are stable but time-consuming, while the existing closed-form solution is faster but less stable. In this paper, we discuss the homography decomposition problem from a different viewpoint. In contrast to the existing methods which focus on the properties of the homography matrix, we propose a new method that uses three random point correspondences to obtain the motion parameters in closed form. The proposed method is conceptually simple, easy to understand and implement, and has a good geometrical interpretation. This solution can be seen as an alternative to the existing closed-form solution. We also discuss the configurations where the closed-form solutions might be unstable and present a framework for homography decomposition taking into account both the efficiency and stability.

Keywords Homography · Decomposition · Relative pose

1 Introduction

Finding the relative pose between two cameras is one of the fundamental vision problems with many applications, for example, in structure-from-motion Schonberger and Frahm (2016), visual localization Sarlin et al. (2019), augmented reality Sattler et al. (2016) and SLAM Mur-Artal and Tardós (2017).

The relative pose between two calibrated cameras, *i.e.*, the relative rotation and translation up to an unknown scale can be extracted from the so-called 3×3 essential matrix, which is a singular matrix with two equal singular values. The essential matrix has 5 degrees of freedom (DoF), and

can be estimated using the well-known 5-point relative pose solver Nistér (2004). The 5-point solver is quite efficient and works for general as well as planar scenes. However, for scenes with a dominant plane, *i.e.*, scenes where most of the detected image points are projections of 3D points lying on a single plane, a more efficient solver is available.

Images of planar scenes or scenes with one dominant plane, *e.g.* facades of buildings, walls, floors, are quite common in indoor and outdoor environments. For planar scenes, the relative pose estimation reduces to the computation of the so-called homography matrix $\mathbf{H}$. In this case, each 2D-2D point correspondence provides two linear homogeneous constraints on the 3×3 homography matrix $\mathbf{H}$. Since the homography matrix can be estimated only up to scale, four point correspondences are sufficient to determine $\mathbf{H}$. The well-known 4-point algorithm for estimating homography Hartley and Zisserman (2003) solves a system of eight homogeneous linear equations. Compared to up to ten solutions for the essential matrix obtained from the 5-point relative pose solver, the 4-point homography solver returns only one solution for $\mathbf{H}$. Assuming calibrated cameras, the relative pose, *i.e.*, the relative rotation and the translation up to scale, can

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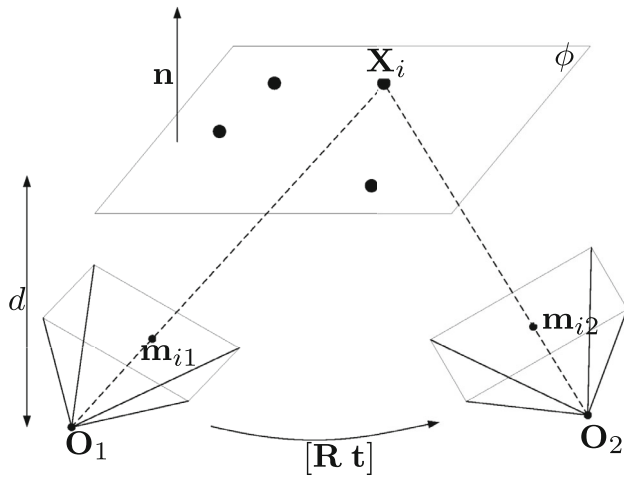


Fig. 1 Points on a plane viewed by two cameras

be extracted from the homography matrix $\mathbf{H}$ using a process known as homography decomposition.

Current methods for homography decomposition fall into two categories: SVD-based methods Faugeras and Lustman (1988), Zhang and Hanson (1996), Ma et al. (2004) and closed-form solutions Malis and Vargas Villanueva (2007), Yang et al. (2018). While state-of-the-art methods are generally fast, stable, and suitable for many applications, there are still scenarios where they encounter limitations. In our extensive experiments, we have observed that SVD-based methods offer better numerical stability compared to closed-form solutions, albeit at the expense of increased computational time. The closed-form solution, while more efficient, may yield inaccurate estimates.

In this paper, we propose a novel approach to homography decomposition, departing from traditional methodologies. Rather than relying on the properties of the homography matrix, we define three points in the first view and establish their corresponding points in the second view. Utilizing the principles of rigid motion, we derive the motion parameters directly in a closed form. This new algorithm involves solving a cubic and a quadratic equation, making it computationally tractable. Moreover, we address the scenarios where the closed-form solution may falter by examining the discriminant of the cubic equation. Finally, to strike a balance between accuracy and efficiency, we introduce a framework that combines the strengths of both the closed-form solution and the SVD-based method for homography decomposition.

The rest of the paper is organized as follows: Sec. 2 establishes some notation and describes the homography decomposition problem. Sec. 2.2 gives a brief review of existing methods for homography decomposition. Sec. 3 presents the proposed method and discusses the failure cases. Sec. 4 gives experimental results. Sec. 5 concludes.

2 Problem Statement

Assume that we have a set of 3D points $\{\mathbf{X}_i, i = 1, \dots, k\}$ observed by two cameras, with projection matrices $\mathbf{P}_1 = \mathbf{K}_1[\mathbf{I}|\mathbf{0}]$ and $\mathbf{P}_2 = \mathbf{K}_2[\mathbf{R}|\mathbf{t}]$. Here, $\mathbf{R} \in SO(3)$ is the unknown rotation and $\mathbf{t} \in \mathbb{R}^3$ the unknown translation between these two cameras. The centers of the first and second cameras are defined as $\mathbf{O}_1, \mathbf{O}_2$, respectively. We assume that the 3×3 calibration matrices $\mathbf{K}_1$ and $\mathbf{K}_2$ of both cameras are known. Let $\{\mathbf{m}_{i1}, \mathbf{m}_{i2}\}, i = 1, 2, 3, \dots, k$ be k image point correspondences in the first and second camera, respectively. In this case, we have the normalized image points $\mathbf{x}_{i2} = \mathbf{K}_2^{-1}\mathbf{m}_{i2}$, and $\mathbf{x}_{i1} = \mathbf{K}_1^{-1}\mathbf{m}_{i1}$, where $\{\mathbf{m}_{i1}, \mathbf{m}_{i2}, \mathbf{x}_{i1}, \mathbf{x}_{i2}\}$ are all represented using homogeneous coordinates. Let $\mathbf{X}_{i1}$ and $\mathbf{X}_{i2}$ be the coordinates of the i^{th} 3D point $\mathbf{X}_i$ in the coordinate system of the first and second camera, respectively. This gives us

$$\mathbf{X}_{i2} = \mathbf{R}\mathbf{X}_{i1} + \mathbf{t}. \quad (1)$$

If all 3D points $\mathbf{X}_i$ lying on a plane ϕ in the 3D space, we have the following equation

$$\mathbf{n}^\top \mathbf{X}_{i1} = d, \quad (2)$$

where $\mathbf{n}$ is the unit normal vector of the plane with respect to the first camera frame, and d denotes the distance from the plane to the optical center of the first camera. We select d as the positive distance, and choose the unit normal $\mathbf{n}$ to point away from the origin. Substituting (2) into (1), we have

$$\mathbf{X}_{i2} = \mathbf{R}\mathbf{X}_{i1} + \frac{\mathbf{t}}{d}\mathbf{n}^\top \mathbf{X}_{i1}. \quad (3)$$

Instead of directly estimating the unknown rotation $\mathbf{R}$ and the unknown translation $\mathbf{t}$, the standard approach is to estimate the unknown 3×3 matrix $\mathbf{H}$ of the form

$$\mathbf{H} = \mathbf{R} + \frac{\mathbf{t}}{d}\mathbf{n}^\top. \quad (4)$$

After substituting (4) into (3) we have

$$\mathbf{X}_{i2} = \mathbf{H}\mathbf{X}_{i1}. \quad (5)$$

The 3×3 matrix $\mathbf{H}$ is the so-called Euclidean homography matrix. Substituting the normalized image points $\mathbf{x}_{i2}, \mathbf{x}_{i1}$ into (5) results in

$$\lambda_{i2}\mathbf{x}_{i2} = \mathbf{H}\lambda_{i1}\mathbf{x}_{i1}, \quad (6)$$

where $\lambda_{i2}, \lambda_{i1}$ are the depths of the image points $\mathbf{x}_{i1}$ and $\mathbf{x}_{i2}$ in each camera's coordinate system. The depths $\lambda_{i2}, \lambda_{i1}$ can be eliminated from equation (6) by multiplying it from the

left with the skew-symmetric matrix $[\mathbf{x}_{i2}]_{\times}$ of the vector $\mathbf{x}_{i2}$. Since $[\mathbf{x}_{i2}]_{\times} \mathbf{x}_{i2} = 0$, equation (6) can be rewritten as

$$[\mathbf{x}_{i2}]_{\times} \mathbf{H} \mathbf{x}_{i1} = 0, \quad (7)$$

Each 2D-2D point correspondence $\mathbf{x}_{i1} \leftrightarrow \mathbf{x}_{i2}$ gives three homogeneous equations of the form (7). However, since the skew-symmetric matrix $[\mathbf{x}_{i2}]_{\times}$ has rank two, only two of these three equations are linearly independent. Thus, to estimate the unknown 3×3 homography matrix $\mathbf{H}$ we need four 2D-2D point correspondences. Using four point correspondences to solve the system of homogenous equations (7), we can obtain the calibrated homography matrix $\tilde{\mathbf{H}}$, which is equivalent to $\mathbf{H}$ up to a scale factor. Euclidean homography $\mathbf{H}$ (4) is a special calibrated homography matrix whose median singular value is 1 Zhang and Hanson (1995). Thus the relationship between $\tilde{\mathbf{H}}$ and $\mathbf{H}$ is defined as

$$\mathbf{H} = \pm \tilde{\mathbf{H}} / \text{median}(\text{svd}(\tilde{\mathbf{H}})) \quad (8)$$

The goal of homography decomposition is to recover $\{\mathbf{R}, \mathbf{t}, \mathbf{n}, d\}$ from a calibrated homography matrix $\tilde{\mathbf{H}}$.

2.1 Chirality Check

Before decomposition, we first need to perform a chirality check. For the standard two-view homography estimation, both cameras should be on the same side of the plane ϕ . Thus, we have

$$\mathbf{n}^T \overrightarrow{\mathbf{O}_1 \mathbf{O}_2} < d, \quad (9)$$

where the projection of $\overrightarrow{\mathbf{O}_1 \mathbf{O}_2}$ in the direction of the normal vector $\mathbf{n}$ is shorter than d (see Fig. 1). Since the center of the second camera $\mathbf{O}_2$ is given by $-\mathbf{R}^T \mathbf{t}$, we have

$$-\mathbf{n}^T \mathbf{R}^T \mathbf{t} < d \Rightarrow 1 + \mathbf{n}^T \mathbf{R}^T \frac{\mathbf{t}}{d} > 0,$$

Based on the matrix determinant lemma Harville (1998), we have

$$\begin{aligned} \det(\mathbf{H}) &= \det\left(\mathbf{R} + \frac{\mathbf{t}}{d} \mathbf{n}^T\right) \\ &= \det(\mathbf{R}) \left(1 + \mathbf{n}^T \mathbf{R}^{-1} \frac{\mathbf{t}}{d}\right) \\ &= 1 + \mathbf{n}^T \mathbf{R}^T \frac{\mathbf{t}}{d}. \end{aligned}$$

Hence, the chirality check is equivalent to checking the sign of the determinant of the Euclidean homography matrix $\mathbf{H}$. In this paper, we use the terms physically possible/impossible homography to refer to Euclidean homographies that can be decomposed into physically possible/impossible motions,

i.e., when two cameras are on the same/opposite side of the plane ϕ .

In practice, $\mathbf{H}$ is usually found by maximizing the number of inliers using the following cost Hartley and Zisserman (2003)

$$\epsilon(\mathbf{x}_{i1}, \mathbf{H}^{-1} \mathbf{x}_{i2}) + \epsilon(\mathbf{x}_{i2}, \mathbf{H} \mathbf{x}_{i1}) < \text{threshold},$$

where $\epsilon(\cdot)$ represents the Euclidean distance between the two inhomogeneous points. If $\mathbf{H}$ is the best homography that maximizes the number of inliers, then $-\mathbf{H}$ can also maximize the number of inliers. In this case, when performing decomposition, if $\mathbf{H}$ is physically impossible ($\det(\mathbf{H}) < 0$), we only need to decompose $-\mathbf{H}$.

2.2 Existing Methods

In this section, we do a brief review of existing methods.

SVD-based methods. Classical methods for homography decomposition are based on SVD. SVD-based method was first introduced in Faugeras and Lustman (1988), and was also discussed in Zhang and Hanson (1996) and Ma et al. (2004). Among the three SVD-based methods, the one illustrated in Ma et al. (2004) is possibly the best choice. It has better geometric meanings and simpler formulation. Hence, we only illustrate the SVD-based method from Section 5.3.3 in Ma et al. (2004). For the other two methods, we refer the reader to Faugeras and Lustman (1988), Zhang and Hanson (1996). The discussed SVD-based method is based on the following theorem

Theorem 1 $\mathbf{H}$ preserves the length of any vector orthogonal to the vector $\mathbf{n}$ of the plane ϕ , i.e., if $\mathbf{n} \perp \mathbf{e}$ for some $\mathbf{e} \in \mathbb{R}^3$, we have $\|\mathbf{H}\mathbf{e}\| = \|\mathbf{R}\mathbf{e}\| = \|\mathbf{e}\|$.

The proof is straightforward since

$$\mathbf{H}\mathbf{e} = \left(\mathbf{R} + \frac{\mathbf{t}}{d} \mathbf{n}^T\right) \mathbf{e} = \mathbf{R}\mathbf{e}.$$

Furthermore, if we know the plane spanned by the vectors that are orthogonal to $\mathbf{n}$, we can determine $\mathbf{n}$ itself. Based on this property, we can first recover the normal vector and then the rotation and translation can be found.

Let $\sigma_1^2 \geq \sigma_2^2 \geq \sigma_3^2 \geq 0$ be three eigenvalues of the symmetric matrix $\mathbf{H}^T \mathbf{H}$. Based on Zhang and Hanson (1995) we know that $\sigma_2 = 1$. Here we assume $\mathbf{H}$ is not a rotation matrix, i.e., $\sigma_1^2 > \sigma_3^2$. Since $\mathbf{H}^T \mathbf{H}$ is symmetric, it can be diagonalized by an orthogonal matrix $\mathbf{V} \in \text{SO}(3)$ such that

$$\mathbf{H}^T \mathbf{H} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T,$$

where $\mathbf{\Lambda} = \text{diag}\{\sigma_1^2, \sigma_2^2, \sigma_3^2\}$, and $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3]$. To ensure that the decomposition gives valid rotation matrices,

we need to check the sign of $\det(\mathbf{V})$: if $\det(\mathbf{V}) < 0$, the sign of $\mathbf{V}$ is reversed. We have

$$\mathbf{H}^\top \mathbf{H} \mathbf{v}_1 = \sigma_1^2 \mathbf{v}_1, \mathbf{H}^\top \mathbf{H} \mathbf{v}_2 = \mathbf{v}_2, \mathbf{H}^\top \mathbf{H} \mathbf{v}_3 = \sigma_3^2 \mathbf{v}_3,$$

where $\mathbf{v}_2 = [\mathbf{n}]_\times \mathbf{R}^\top \mathbf{t}$, and its length is preserved under the map $\mathbf{H}$. The vector $\mathbf{v}_2 = [\mathbf{n}]_\times \mathbf{R}^\top \mathbf{t} = \mathbf{n} \times (\mathbf{R}^\top \mathbf{t})$ represents a triple product that is inherently orthogonal to both $\mathbf{n}$ and $\mathbf{R}^\top \mathbf{t}$, satisfying $\mathbf{n}^\top \mathbf{v}_2 = 0$ and $(\mathbf{R}^\top \mathbf{t})^\top \mathbf{v}_2 = 0$. In addition, we define two unit-length vectors that are orthogonal to $\mathbf{v}_2$ as

$$\mathbf{u}_{1,2} = \frac{\sqrt{1 - \sigma_3^2} \mathbf{v}_1 \pm \sqrt{\sigma_1^2 - 1} \mathbf{v}_3}{\sqrt{\sigma_1^2 - \sigma_3^2}},$$

The lengths of $\mathbf{u}_1$ and $\mathbf{u}_2$ are also preserved under the map $\mathbf{H}$. We can see that $\mathbf{H}$ preserves the length of all vectors from each of the two subspaces

$$S_1 = \text{span}\{\mathbf{v}_2, \mathbf{u}_1\}, S_2 = \text{span}\{\mathbf{v}_2, \mathbf{u}_2\}.$$

Since $\mathbf{v}_2$ is orthogonal to $\mathbf{u}_1$ and $\mathbf{u}_2$, $\mathbf{v}_2 \times \mathbf{u}_1$ is a unit normal vector to S_1 , and $\mathbf{v}_2 \times \mathbf{u}_2$ a unit normal vector to S_2 . Then $\{\mathbf{v}_2, \mathbf{u}_1, \mathbf{v}_2 \times \mathbf{u}_1\}$ and $\{\mathbf{v}_2, \mathbf{u}_2, \mathbf{v}_2 \times \mathbf{u}_2\}$ form two sets of orthonormal bases for $\mathbb{R}^3$. If $\mathbf{n}$ is the normal to the subspace $S_i, i = 1, 2$, we have

$$\mathbf{R} \mathbf{v}_2 = \mathbf{H} \mathbf{v}_2, \mathbf{R} \mathbf{u}_i = \mathbf{H} \mathbf{u}_i, \mathbf{R} \mathbf{v}_2 \times \mathbf{R} \mathbf{u}_i = \mathbf{H} \mathbf{v}_2 \times \mathbf{H} \mathbf{u}_i.$$

We define the following matrices

$$\mathbf{U}_1 = [\mathbf{v}_2, \mathbf{u}_1, \mathbf{v}_2 \times \mathbf{u}_1], \mathbf{W}_1 = [\mathbf{H} \mathbf{v}_2, \mathbf{H} \mathbf{u}_1, \mathbf{H} \mathbf{v}_2 \times \mathbf{H} \mathbf{u}_1], \\ \mathbf{U}_2 = [\mathbf{v}_2, \mathbf{u}_2, \mathbf{v}_2 \times \mathbf{u}_2], \mathbf{W}_2 = [\mathbf{H} \mathbf{v}_2, \mathbf{H} \mathbf{u}_2, \mathbf{H} \mathbf{v}_2 \times \mathbf{H} \mathbf{u}_2].$$

Then the motion parameters are given by

$$\mathbf{n}_1 = \pm \mathbf{v}_2 \times \mathbf{u}_1, \mathbf{R}_1 = \mathbf{W}_1 \mathbf{U}_1^\top, \frac{\mathbf{t}_1}{d_1} = (\mathbf{H} - \mathbf{R}_1) \mathbf{n}_1, \\ \mathbf{n}_2 = \pm \mathbf{v}_2 \times \mathbf{u}_2, \mathbf{R}_2 = \mathbf{W}_2 \mathbf{U}_2^\top, \frac{\mathbf{t}_2}{d_2} = (\mathbf{H} - \mathbf{R}_2) \mathbf{n}_2.$$

In general, we may have four possible solutions to the geometric parameters $\{\mathbf{R}, \mathbf{t}, \mathbf{n}\}$.

Closed-form solution. The closed-form method proposed by Malis and Vargas Malis and Vargas Villanueva (2007) offers analytical solutions for the problem, departing from conventional numerical approaches. Its primary advantage lies in providing a deeper understanding of the homography decomposition problem. For instance, it enables the extraction of relationship among potential solutions, thereby enhancing insights into the problem domain. The paper Malis and Vargas Villanueva (2007) proposed two different variants of the closed-form method: one computes the normal vector first

(using $\mathbf{H}^\top \mathbf{H}$), while the other computes the translation first (using $\mathbf{H} \mathbf{H}^\top$). They are mathematically equivalent, and we refer the reader to Malis and Vargas Villanueva (2007) for more details. Note that all methods for homography decomposition can have two variants, *i.e.*, one variant that uses the decomposition of $\mathbf{H}$, which gives the normal vector first, and the other variant that uses the decomposition of $\mathbf{H}^\top$, which gives the translation vector first. There is no difference between the two variants, since $\mathbf{H}^\top$ can be considered as a new $\mathbf{H}$. In this paper, we evaluate only the variant that computes the normal vector first. In Yang et al. (2018), the authors analyze the properties of the matrix $\mathbf{H} \mathbf{H}^\top$ and demonstrate that the translation vector can be directly computed from its elements. In addition, the authors discuss all the possible degenerate cases based on analyzing the elements of the matrix $\mathbf{H} \mathbf{H}^\top$ and the translation vector.

Although both the closed-form methods from Malis and Vargas Villanueva (2007) and Yang et al. (2018) are efficient and provide better understanding of the problem than the SVD-based method, they are less stable than the SVD-based decomposition. Our aim is to propose a solution that is both fast and stable.

3 Our Approach

In this section, we present the derivation of the proposed method. In particular, given a calibrated homography matrix $\tilde{\mathbf{H}}$, we show that the scale factor can be fixed by solving for one root of a cubic. Then by generating three special points in the 3D space, we show that the location of the 3D points can be recovered, and the motion parameters can be found from 3D-3D point correspondences. Finally, we discuss the failure cases and propose a fast and stable framework to homography decomposition by combining the proposed method and the SVD technique.

3.1 Fixing the Scale Factor

Given $\tilde{\mathbf{H}}$ (we assume $\tilde{\mathbf{H}}$ is not a rotational homography), we first need to fix the scale factor to obtain the Euclidean homography $\mathbf{H}$. To find the scale factor (median singular value of $\tilde{\mathbf{H}}$), the characteristic polynomial equation

$$\det(\tilde{\mathbf{H}}^\top \tilde{\mathbf{H}} - \gamma \mathbf{I}) = 0. \quad (10)$$

is used to compute the eigenvalues of $\tilde{\mathbf{H}}^\top \tilde{\mathbf{H}}$. The characteristic polynomial equation (10) results in a polynomial of degree three in γ

$$\gamma^3 + k_2 \gamma^2 + k_1 \gamma + k_0 = 0, \quad (11)$$

with

$$\begin{aligned} k_2 &= -(c_{11} + c_{22} + c_{33}), \\ k_1 &= c_{11}c_{22} + c_{11}c_{33} + c_{22}c_{33} + \dots \\ &\quad - 2(c_{12}^2 + c_{13}^2 + c_{23}^2), \\ k_0 &= c_{12}^2c_{33} + c_{13}^2c_{22} + c_{23}^2c_{11} + \dots \\ &\quad - c_{11}c_{22}c_{33} - 2c_{12}c_{13}c_{23}, \end{aligned} \quad (12)$$

where c_{ij} are the elements of the symmetric matrix $\mathbf{C} = \tilde{\mathbf{H}}^\top \tilde{\mathbf{H}}$, which has three non-negative real eigenvalues. To find the roots, we use the trigonometric solution Zwillinger (2018). Let

$$\begin{aligned} \alpha &= \frac{3k_1 - k_2^2}{9}, \\ \beta &= \frac{9k_2k_1 - 27k_0 - 2k_2^3}{54}. \end{aligned} \quad (13)$$

Since $\mathbf{C}$ has three real eigenvalues, we have $\alpha < 0$ based on the cubic root formula Zwillinger (2018). Let

$$\Delta = \frac{\beta}{\sqrt{-\alpha^3}} \quad (14)$$

be the discriminant and let $\cos \theta = \Delta$, i.e., $\theta = \arccos(\Delta)$, $\theta \in [0, \pi]$. Now the three roots of (11) can be found using the trigonometric solution Zwillinger (2018) as

$$\begin{aligned} \gamma_1 &= -\frac{k_2}{3} + 2\sqrt{-\alpha} \cos\left(\frac{\theta}{3}\right), \\ \gamma_2 &= -\frac{k_2}{3} + 2\sqrt{-\alpha} \cos\left(\frac{\theta}{3} - \frac{2\pi}{3}\right), \\ \gamma_3 &= -\frac{k_2}{3} + 2\sqrt{-\alpha} \cos\left(\frac{\theta}{3} + \frac{2\pi}{3}\right). \end{aligned} \quad (15)$$

We have

$$\begin{aligned} \gamma_1 - \gamma_2 &= 2\sqrt{-\alpha} \left(-2 \sin \frac{\theta - \pi}{3} \sin \frac{\pi}{3} \right) \geq 0, \\ \gamma_2 - \gamma_3 &= 2\sqrt{-\alpha} \left(-2 \sin \frac{\theta}{3} \sin \frac{-2\pi}{3} \right) \geq 0. \end{aligned} \quad (16)$$

Hence γ_2 is the median eigenvalue of $\mathbf{C}$, and in practice we only need to compute γ_2 . The Euclidean homography is given by

$$\mathbf{H} = \frac{\tilde{\mathbf{H}}}{\sqrt{\gamma_2}}. \quad (17)$$

Note that, the median root of the cubic equation (11) has also been discussed in Malis and Vargas Villanueva (2007). The authors analyzed the roots of the cubic equation using

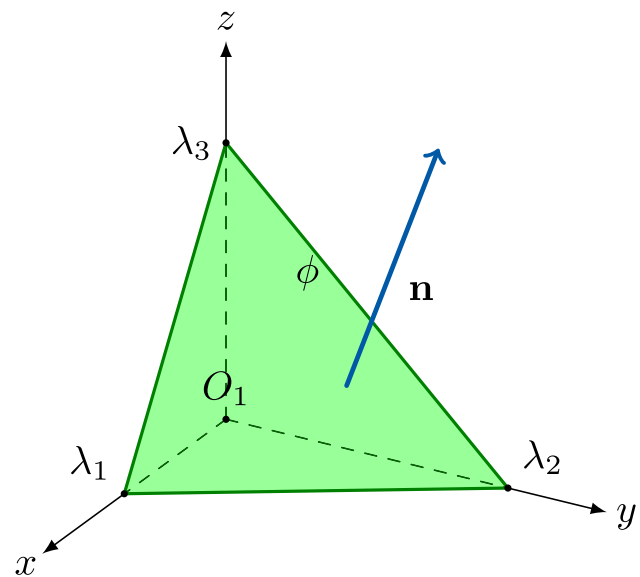


Fig. 2 Three 3D points in the first camera coordinate

Cardano's formula Abramowitz and Stegun (1968), which needs to deal with complex numbers.

3.2 Generating Point Correspondences

The homography $\mathbf{H}$ that we are decomposing is induced by the plane ϕ . This plane intersects the three axes of the coordinate system of the first camera in three points. Let these points be $\mathbf{X}_{11} = [\lambda_1, 0, 0]^\top$, $\mathbf{X}_{21} = [0, \lambda_2, 0]^\top$, $\mathbf{X}_{31} = [0, 0, \lambda_3]^\top$ (See Fig. 2). Let

$$\mathbf{H} = \begin{bmatrix} h_1 & h_2 & h_3 \\ h_4 & h_5 & h_6 \\ h_7 & h_8 & h_9 \end{bmatrix}. \quad (18)$$

Since the corresponding 3D points in the coordinate system of the second camera are given by $\mathbf{X}_{i2} = \mathbf{H}\mathbf{X}_{i1}$, we have

$$\begin{aligned} \mathbf{X}_{12} &= \lambda_1[h_1, h_4, h_7]^\top, \\ \mathbf{X}_{22} &= \lambda_2[h_2, h_5, h_8]^\top, \\ \mathbf{X}_{32} &= \lambda_3[h_3, h_6, h_9]^\top. \end{aligned} \quad (19)$$

Due to the rigid motion, the distance between the two points should be maintained. We have

$$\begin{aligned} \|\mathbf{X}_{12} - \mathbf{X}_{22}\| &= \|\mathbf{X}_{11} - \mathbf{X}_{21}\|, \\ \|\mathbf{X}_{12} - \mathbf{X}_{32}\| &= \|\mathbf{X}_{11} - \mathbf{X}_{31}\|, \\ \|\mathbf{X}_{22} - \mathbf{X}_{32}\| &= \|\mathbf{X}_{21} - \mathbf{X}_{31}\|. \end{aligned} \quad (20)$$

Equations (20) are homogenous in $\{\lambda_1, \lambda_2, \lambda_3\}$. Thus, without loss of generality, we can set $\lambda_3 = 1$ (the plane in the first camera coordinate can not pass through the origin, which

leads to $d = 0$), and obtain three quadratics in two unknowns $\{\lambda_1, \lambda_2\}$. Note that they are not over-constrained. Two distinct quadratics in two unknowns have four solutions, and in our case, three quadratics reduce the number of solutions to two. Squaring both sides in the last equation of (20) results in a quadratic equation in λ_2

$$a_2 \lambda_2^2 + a_1 \lambda_2 + a_0 = 0, \quad (21)$$

where

$$\begin{aligned} a_2 &= h_2^2 + h_5^2 + h_8^2 - 1, \\ a_1 &= -2h_2h_3 - 2h_5h_6 - 2h_8h_9, \\ a_0 &= h_3^2 + h_6^2 + h_9^2 - 1. \end{aligned} \quad (22)$$

By combining the first two equations of (20), so that λ_1^2 can be eliminated and λ_1 is linear in λ_2 . We get

$$b_1 \lambda_1 + b_0 = 0, \quad (23)$$

where

$$\begin{aligned} b_1 &= (h_1h_2 + h_4h_5 + h_7h_8)\lambda_2 - h_1h_3 - h_4h_6 - h_7h_9, \\ b_0 &= a_0 + \frac{1}{2}\lambda_2 a_1. \end{aligned}$$

In general, we first solve (21) to get the solutions to λ_2 , then λ_1 can be found using (23). There are up to two possible solutions to $\{\lambda_1, \lambda_2\}$. Note that since λ_1, λ_2 have geometric interpretations as the intersections of the homography-induced plane with the coordinate axes, real solutions are guaranteed to exist.

3.3 Gauss-Newton Refinement

Although in most cases the above-mentioned method for recovering the median singular value of $\tilde{\mathbf{H}}$ and the values of λ_1 and λ_2 returns solutions with sufficient accuracy, small errors in these estimates can sometimes propagate to larger errors in the recovered motion parameters. To address this, we apply Gauss-Newton refinement to improve the accuracy of the depth estimates, incorporating the median singular value in the process. This refinement can enhance the precision of the estimates. Once λ_i are known, we can perform Gauss-Newton optimization on the sum of squares of (20) to refine the solutions. We have two options on how to do the refinement.

Refine $\{\lambda_1, \lambda_2\}$. A straightforward way is to refine only the two depth parameters $\{\lambda_1, \lambda_2\}$. This can be done using the first two equations of (20), or all three equations from (20).

Refine $\{\lambda_1, \lambda_2\}$ and the scale factor. Another option is to take into account the error in the estimated scale factor $\sqrt{\gamma_2}$

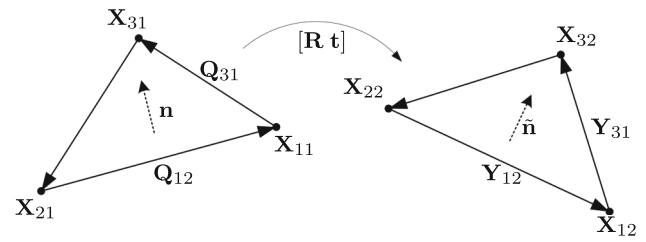


Fig. 3 The rotation $\mathbf{R}$ and translation $\mathbf{t}$ are obtained by registering three 3D points in the two camera coordinates

in (17). Due to the floating-point round-off errors, the homography $\mathbf{H}$ estimated using (17) may not have the median singular value equal to 1. This means that $\mathbf{H} = \sqrt{s}\mathbf{H}_g$, for some scalar value $\sqrt{s}$ and the ground-truth Euclidean homography $\mathbf{H}_g$. We found that refining s can further improve the numerical stability while maintaining efficiency (see Sec. 4.2 for details). To refine both s and depths $\{\lambda_1, \lambda_2\}$ we use three equations in (20) in the form

$$\begin{aligned} \|\mathbf{X}_{12} - \mathbf{X}_{22}\| &= \sqrt{s}\|\mathbf{X}_{11} - \mathbf{X}_{21}\|, \\ \|\mathbf{X}_{12} - \mathbf{X}_{32}\| &= \sqrt{s}\|\mathbf{X}_{11} - \mathbf{X}_{31}\|, \\ \|\mathbf{X}_{22} - \mathbf{X}_{32}\| &= \sqrt{s}\|\mathbf{X}_{21} - \mathbf{X}_{31}\|. \end{aligned} \quad (24)$$

By squaring both sides of (24), we can compute the Jacobian matrix and apply the Gauss-Newton refinement. The initial value of s is set to 1.

3.4 Motion Recovery

Once we have $\{\lambda_1, \lambda_2\}$, we have three 3D points in two coordinate systems, *i.e.*, the coordinate system of the first and the coordinate system of the second camera. The situation is illustrated in Fig. 3. Let

$$\begin{aligned} \mathbf{Y}_{12} &= \mathbf{X}_{12} - \mathbf{X}_{22}, \quad \mathbf{Y}_{31} = \mathbf{X}_{32} - \mathbf{X}_{12}, \\ \mathbf{Q}_{12} &= \mathbf{X}_{11} - \mathbf{X}_{21}, \quad \mathbf{Q}_{31} = \mathbf{X}_{31} - \mathbf{X}_{11}. \end{aligned} \quad (25)$$

Note that while in $\mathbf{X}_{ij}$ the first index indicates the point and the second index the camera, in $\mathbf{Y}_{ij}$ and $\mathbf{Q}_{ij}$ both indices refer to points, *i.e.*, two points that were used to generate points $\mathbf{Y}_{ij}$ and $\mathbf{Q}_{ij}$. Here, $\mathbf{Q}_{ij}$ are points in the coordinate system of the first camera and $\mathbf{Y}_{ij}$ are points in the coordinate system of the second camera. Two pairs of solutions for the normal of the plane ϕ in the first and second camera coordinates are given by (see Fig. 3)

$$\mathbf{n} = \mathbf{Q}_{12} \times \mathbf{Q}_{31}, \quad \tilde{\mathbf{n}} = \mathbf{Y}_{12} \times \mathbf{Y}_{31} \quad (26)$$

Based on (2), the distance d between the center of the first camera and the plane ϕ is given by

$$d = \mathbf{n}^\top \mathbf{X}_{11}. \quad (27)$$

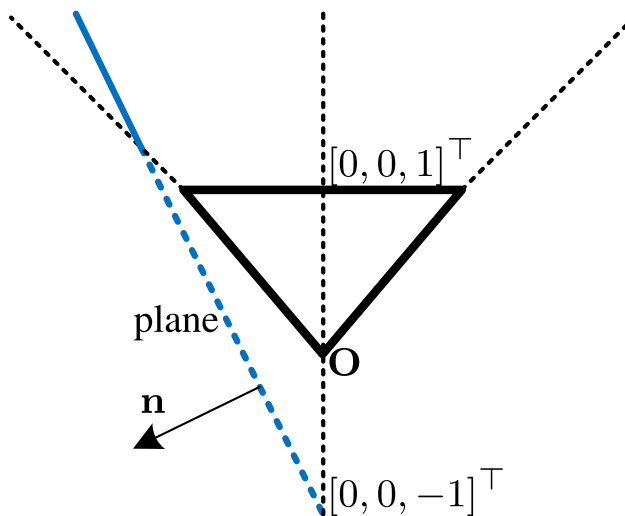


Fig. 4 The plane may intersect the z -axis at $[0, 0, -1]^T$

If $d < 0$, the signs of $\mathbf{n}$ and d are reversed. The distance d here is calculated by scaling the 3D points so that the plane intersects the z -axis at $[0, 0, 1]^T$, *i.e.*, by fixing $\lambda_3 = 1$ in our computations. In general, it is not possible to recover the actual distance d due to the scale ambiguity. On the other hand, we have $\tilde{\mathbf{n}} = \mathbf{R}\mathbf{n}$, thus we obtain

$$\mathbf{Y} = \mathbf{R}\mathbf{Q}, \quad (28)$$

with

$$\mathbf{Y} = [\mathbf{Y}_{12}, \mathbf{Y}_{31}, \tilde{\mathbf{n}}], \quad \mathbf{Q} = [\mathbf{Q}_{12}, \mathbf{Q}_{31}, \mathbf{n}]. \quad (29)$$

Since $\mathbf{Q}_{12}, \mathbf{Q}_{31}, \mathbf{Q}_{12} \times \mathbf{Q}_{31}$ are not coplanar, *i.e.*, the matrix $\mathbf{Q}$ is invertible. The rotation and translation are given by

$$\begin{aligned} \mathbf{R} &= \mathbf{Y}\mathbf{Q}^{-1}, \\ \mathbf{t} &= \mathbf{X}_{12} - \mathbf{R}\mathbf{X}_{11}. \end{aligned} \quad (30)$$

Note that, the plane may intersect the z -axis at $[0, 0, -1]^T$ (see Fig. 4). In this scenario, the geometric parameters are represented as $-\mathbf{n}, \mathbf{R}, -\mathbf{t}$. While Ma et al. (2004) (Page 137) claimed that only two physically possible solutions exist under the assumption that the plane always passes through $[0, 0, 1]^T$, Fig. 4 illustrates that the plane can also pass through $[0, 0, -1]^T$. In general, there are up to four physically possible solutions: two possible solutions to λ_1, λ_2 , and each λ_1, λ_2 gives two possible motion parameters.

3.5 Failure Cases

Although the method presented in Sections 3.1–3.4 returns accurate solutions in almost all scenarios, we have found that it may still fail for some special corner cases. Based on the

experiments, we have observed that the failure cases in our method correlate with the value of the discriminant Δ (14). When Δ approaches 1, the proposed method becomes unstable (see Sec. 4.2 for details). To address this issue, we propose a simple check in which we evaluate Δ against a given threshold τ . If $\Delta < \tau$, we utilize the proposed method; conversely, if $\Delta \geq \tau$, we switch to the SVD-based method that is described in Sec. 2.2. Through experiments, we have found that $\tau = 0.99999$ yields good results in terms of speed and accuracy. Additionally, if $\tilde{\mathbf{H}}^\top \tilde{\mathbf{H}}$ is a scaled identity matrix, then $\tilde{\mathbf{H}}$ is a rotational homography.

3.6 Special Target Planes

The target plane may not always intersect all three axes of the coordinate system of the first camera. We summarize such special cases in the following table.

n_x	n_y	n_z	Properties
1	0	0	$a_2 = a_1 = a_0 = 0, \lambda_1 \rightarrow 0$
0	1	0	$a_0 = 0, b_1 = b_0 = 0, \lambda_2 \rightarrow 0$
0	0	1	$a_2 = 0, \lambda_2 \rightarrow +\infty, \lambda_1 \rightarrow +\infty$
0	-	-	$b_1 = 0, \lambda_1 \rightarrow +\infty$
-	0	-	$a_2 = 0, \lambda_2 \rightarrow +\infty$
-	-	0	$a_0 = 0, \lambda_2 \rightarrow 0, \lambda_1 \rightarrow 0$

In these scenarios, the proposed solver can fail. Such cases rarely appear in practice; however, even if they appear, we can easily avoid them. We can do this by multiplying $\mathbf{H}$ by a random rotation matrix $\mathbf{R}_{ran}$ if such cases are detected based on the values of a_2, a_0, b_1 . Let's consider $\mathbf{H}_r = \mathbf{H}\mathbf{R}_{ran}$, which is transformed by multiplying $\mathbf{H}$ from the right by a random rotation matrix $\mathbf{R}_{ran}$. Then we have the following decomposition of $\mathbf{H}_r$

$$\begin{aligned} \mathbf{H}_r &= \mathbf{R}\mathbf{R}_{ran} + \frac{\mathbf{t}}{d}(\mathbf{R}_{ran}^\top \mathbf{n})^\top, \\ &= \mathbf{R}_r + \frac{\mathbf{t}}{d}\mathbf{n}_r^\top. \end{aligned}$$

We can see that the matrix $\mathbf{R}_{ran}$ can rotate the normal vector $\mathbf{n}$ to avoid zero entries. The original motion parameters can be recovered from $\mathbf{R} = \mathbf{R}_r\mathbf{R}_{ran}^\top$ and $\mathbf{n} = \mathbf{R}_{ran}\mathbf{n}_r$. The whole $\mathbf{H}$ decomposition procedure that we proposed in Sec. 3 is summarized in Algorithm 1.

3.7 Recovering $\mathbf{E}$ from $\mathbf{H}$

In some applications, we may be interested in recovering the essential matrix from a homography matrix. Since the essential matrix is defined as $\mathbf{E} = [\mathbf{t}]_\times \mathbf{R}$, we can recover $\mathbf{E}$ once we get $\mathbf{t}$ and $\mathbf{R}$ from the homography decomposition. How-

Algorithm 1: Homography Decomposition

Input: Calibrated Homography matrix $\tilde{\mathbf{H}}$
Output: Motion parameters $\{\mathbf{R}, \mathbf{t}, \mathbf{n}\}$

```

1 if  $\det(\tilde{\mathbf{H}}) < 0$  then
2    $\tilde{\mathbf{H}} = -\tilde{\mathbf{H}}$ 
3 if  $\tilde{\mathbf{H}}$  is a rational homography then
4    $\mathbf{R} = \tilde{\mathbf{H}}/\text{scale}$ ,  $\mathbf{n} = \mathbf{0}$ ,  $\mathbf{t} = \mathbf{0}$ 
5 Compute  $\alpha, \beta, \Delta$  based on (13)
6 if  $\Delta < 0.99999$  then
7   Compute  $\gamma_2$  based on (15)
8   Fix the scale using (17)
9   Compute  $a_2, a_1, a_0$  based on (22)
10  while  $\text{abs}(a_2)$  or  $\text{abs}(a_0) < 10^{-8}$  do
11    Multiply  $\mathbf{H}$  by a random rotation matrix  $\mathbf{R}_r$ 
12  Compute  $\lambda_2$  (up to two solutions) using (21)
13  Compute  $b_1, b_0$  based on (24)
14  while  $\text{abs}(b_1) < 10^{-8}$  do
15    Multiply  $\mathbf{H}$  by a random rotation matrix  $\mathbf{R}_r$ 
16  Compute  $\lambda_1$  using (23)
17  Compute  $\mathbf{X}_{12}, \mathbf{X}_{22}, \mathbf{X}_{32}$  using (19)
18  Gauss-Newton refinement
19  Compute  $\mathbf{n}, d$  using (26) and (27)
20  if  $d < 0$  then
21     $\mathbf{n} = -\mathbf{n}$ ,  $d = -d$ 
22  Compute  $\mathbf{Y}_{12}, \mathbf{Y}_{31}, \mathbf{Q}_{12}, \mathbf{Q}_{31}$  using (25)
23  Construct matrix  $\mathbf{Y}, \mathbf{Q}$  using (29)
24  Compute  $\mathbf{R}, \mathbf{t}$  using (30)
25 else
26   Using the SVD-based decomposition

```

ever, we also have $\mathbf{E} = [\mathbf{t}]_{\times} \mathbf{H}$, where we can recover $\mathbf{E}$ using the translation $\mathbf{t}$ and the homography $\mathbf{H}$. If we are only interested in the essential matrix recovery, *i.e.*, we don't need the normal vector and rotation matrix, we only need to extract the translation vector. The method illustrated in previous sections first extracts the normal of the plane, then the rotation matrix and finally the translation. Now we show a simple trick to get the translation first. The transpose of $\mathbf{H}$ is given by

$$\mathbf{H}^T = \mathbf{R}^T + \frac{\mathbf{n}}{d} \mathbf{t}^T. \quad (31)$$

Similarly to (4), Eq. (31) also defines a homography. However, in this case $\mathbf{n}$ and $\mathbf{t}$ swap their roles, *i.e.*, the normal vector to the plane that generates this homography $\mathbf{H}^T$ is $\mathbf{t}$, while $\mathbf{n}$ represents the translation. Hence, we can decompose $\mathbf{H}^T$ to get the normal vector, which is the translation corresponding to $\mathbf{H}$. In short, we perform the same steps on $\mathbf{H}^T$ as presented in Sec. 3. Then we terminate at (26), and the essential matrix is then obtained from $[\mathbf{t}]_{\times} \mathbf{H}$.

4 Experiments

In this section, we compare the numerical stability and efficiency of the proposed method with those of state-of-the-art methods. All solvers are implemented in C++ and the experiments are run on a desktop computer under Ubuntu 20.04 with an Intel Core i7-11700K CPU. We compare our decomposition method with methods whose C++ implementations are publicly available in OpenCV: the closed-form solution by Malis and Vargas Malis and Vargas Villanueva (2007)¹ (denoted as Malis and Vargas (O)) and the SVD-based method Zhang and Hanson (1996). Original OpenCV implementations of the state-of-the-art methods are not the most efficient. Thus, for a fair comparison, we also tested modified versions of these methods that use the derivations proposed in this paper. First, the OpenCV implementation of the closed form solution by Malis and Vargas Malis and Vargas Villanueva (2007) uses SVD to obtain the median singular value that is used to fix the scale factor of the homography matrix. However, this goes against the main idea of this closed-form method since the results of such SVD can be directly used for the decomposition. Therefore, for the closed-form solution by Malis and Vargas Malis and Vargas Villanueva (2007), we use (15) to get the median singular value. We denote this version as Malis and Vargas (C), where C represents closed-form. The implementation of SVD-based method performs two SVD decompositions in OpenCV. However, it is evident that the SVD of $\tilde{\mathbf{H}}, \tilde{\mathbf{H}}^T \tilde{\mathbf{H}}$ and $\mathbf{H}$ share the same matrix $\mathbf{V}$, and the only difference between them lies in their singular values. Thus these two SVD decompositions can be replaced by a single one. Therefore, we only test the variants, where we implement the method of Ma et al. (2004) using a single SVD decomposition. The SVD-based methods are denoted as SVD ($\tilde{\mathbf{H}}$) and SVD ($\tilde{\mathbf{H}}^T \tilde{\mathbf{H}}$). In addition, we also implement the method of Yang et al. (2018) based on Eigen library for comparison.

To compute the rotation and translation errors, the most widely used formulation is based on the arccos function. However, as shown in Ding et al. (2023), the precision of arccos formula is limited for errors smaller than 10^{-7} . We thus use the arcsin formula defined in Ding et al. (2023):

$$\xi_{\mathbf{R}} = 2 \arcsin \left(\frac{\|\mathbf{R}_g - \mathbf{R}_e\|}{2\sqrt{2}} \right),$$

$$\xi_{\mathbf{t}} = 2 \arcsin \left(\frac{1}{2} \left\| \frac{\mathbf{t}_e}{\|\mathbf{t}_e\|} - \frac{\mathbf{t}_g}{\|\mathbf{t}_g\|} \right\| \right),$$

where $\mathbf{R}_g, \mathbf{t}_g$ and $\mathbf{R}_e, \mathbf{t}_e$ are the ground truth and the estimated rotation and translation, respectively. The arcsin relationship between two rotation matrices has also been shown in Huynh

¹ [OpenCV:homography_decomp.cpp](#)

Table 1 Comparison of number of failure cases for 10^6 trials. For the closed-form methods, we also provide the statistics of the median singular value $\sqrt{\gamma_2}$ and the discriminant Δ for the failure cases

Method	Failure	$\sqrt{\gamma_2}$			Δ			Method	Failure
		Min	Median	Max	Min	Median	Max		
Ours	1223	$6.54 \cdot 10^{-7}$	$5.0 \cdot 10^{-4}$	0.44	0.9984	$1 - 10^{-13}$	1.0	Malis and Vargas (O)	6
Ours (Refine I)	1071	$6.54 \cdot 10^{-7}$	$4.2 \cdot 10^{-4}$	0.039	$1 - 10^{-9}$	$1 - 10^{-13}$	1.0	SVD ($\tilde{\mathbf{H}}$)	0
Ours (Refine II)	858	$6.54 \cdot 10^{-7}$	$3.3 \cdot 10^{-4}$	0.039	$1 - 10^{-11}$	$1 - 10^{-13}$	1.0	SVD ($\tilde{\mathbf{H}}^T \tilde{\mathbf{H}}$)	0
Ours (Refine s, λ)	764	$6.54 \cdot 10^{-7}$	$3.0 \cdot 10^{-4}$	0.0025	$1 - 10^{-12}$	$1 - 10^{-13}$	1.0		
Malis and Vargas (C)	1534	$6.54 \cdot 10^{-7}$	$5.4 \cdot 10^{-4}$	0.70	0.9881	$1 - 10^{-13}$	1.0		
Yang <i>et al.</i>	1502	$6.54 \cdot 10^{-7}$	$5.3 \cdot 10^{-4}$	0.69	0.9883	$1 - 10^{-13}$	1.0		

(2009). Since there are multiple potential solutions, we evaluate them all and choose the one that gives the smallest error.

4.1 Experimental Settings

To test the solvers, we generated 10^6 scenes in this experiment, with the aim of highlighting potential failure cases. Failure cases are defined as cases where either the rotation error or the translation error exceeds 0.01° . Synthetic data are generated as follows: the unit quaternion is drawn from an isotropic Gaussian distribution and then converted to the ground truth rotation matrix $\mathbf{R}_g$. The ground truth translation $\mathbf{t}_g$ is generated from the standard normal distribution. We randomly generate three non-collinear 3D points $\{\mathbf{X}_{11}, \mathbf{X}_{21}, \mathbf{X}_{31}\}$ in the first camera coordinate, then the corresponding 3D points in the second camera coordinate are given by $\mathbf{X}_{i2} = \mathbf{R}_g \mathbf{X}_{i1} + \mathbf{t}_g$. The ground truth $\mathbf{H}_g$ is obtained based on (5)

$$\mathbf{H}_g = [\mathbf{X}_{12}, \mathbf{X}_{22}, \mathbf{X}_{32}] \cdot [\mathbf{X}_{11}, \mathbf{X}_{21}, \mathbf{X}_{31}]^{-1}. \quad (32)$$

Then we check the sign of $\det(\mathbf{H})$ to ensure the homography matrices have positive determinant. The scaled homography matrix for decomposition is obtained by $\tilde{\mathbf{H}} = \mathbf{H}_g / \|\mathbf{H}_g\|_F$. In this case, the scale of $\tilde{\mathbf{H}}$ is fixed in a similar way as it is in the standard 4-point algorithm for homography estimation. The homography matrix estimated using four 2D-2D point correspondences is usually scaled so that $\|\tilde{\mathbf{H}}\|_F = 1$.

We omit the results with real data, as the differences in accuracy among the various methods are almost negligible in real applications when using RANSAC Fischler and Bolles (1981) with local optimization. The difference lies in the running time, which will be demonstrated in synthetic tests.

4.2 Solver Stability

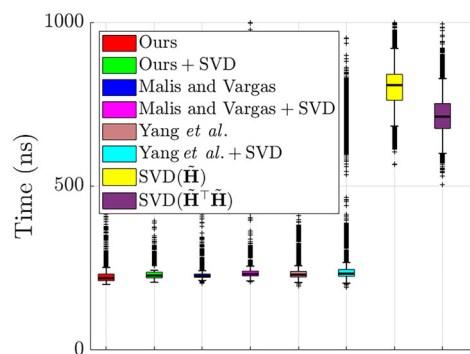
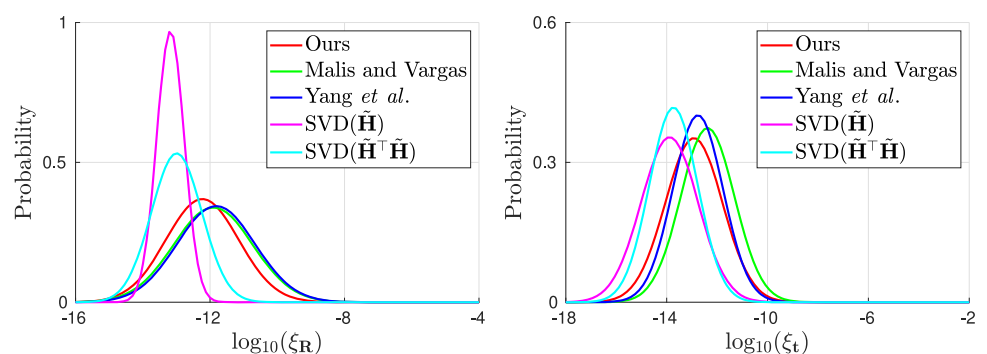
In the first experiment, we measure the number of failure cases of different methods. In order to show the statics for the discriminant Δ , here we run our method without switch-

ing to SVD. Ours (Refine I) and Ours (Refine II) denote the proposed method that uses Gauss-Newton to refine λ_1, λ_2 using the first two equations and all three equations from (20), respectively. Ours (Refine s, λ) denotes the proposed method that uses Gauss-Newton to refine both the scale factor s and λ_1, λ_2 . In Tab. 1, we show the statistics of failure cases. In Tab. 2, we report the mean, standard deviation, and the box plot of the timings, obtained by combining SVD with discriminant-based checking. Our observations indicate that all closed-form solutions exhibit some instabilities. The proposed method is more stable and efficient than the existing closed-form methods, but it still encounters occasional failure cases. Malis and Vargas (O) reported is the right part of Tab. 1, which uses SVD to find the median singular value, is more stable than the closed-form methods. However, as shown in Tab. 2, it is even more time-consuming than SVD-based methods, making it impractical. Since the only difference between Malis and Vargas (C) and Malis and Vargas (O) lies in the method used to fix the scale of the homography matrix, this indicates that the primary cause of instability in the closed-form methods is the computation of the median singular value.

In contrast, SVD-based methods are remarkably stable, with no failure cases detected. However, they are 4-7x slower than the proposed method. With Gauss-Newton refinement, the stability of the proposed method can be slightly improved. In Tab. 1 we also show the statistics of the median singular value $\sqrt{\gamma_2}$ and the discriminant Δ (14) for the failure cases of the closed-form methods. It can be seen that the median singular value $\sqrt{\gamma_2}$ for all failure cases is close to 0. Thus, even a very small error in the median singular value $\sqrt{\gamma_2}$ can result in a large error in the Euclidean homography matrix $\frac{\tilde{\mathbf{H}}}{\sqrt{\gamma_2}}$. Additionally, the discriminant of all failure cases is very close to 1. This can be explained as follows: If the median singular value is close to 0, the smallest singular value should also be close to 0, indicating that the cubic equation has two almost equal roots where we have $\Delta \approx 1$. This suggests that the discriminant check proposed in Section 3.5 with a switch to the SVD-based method if $\Delta \geq \tau$ can eliminate such failure

Table 2 Comparison of speed of the homography decomposition methods for 10^6 trials. The mean, standard deviation, and the box plot of the timings are reported

Method	Time (ns)	
	Mean	Std.
Ours (Refine s, λ)	206	70
Ours (Refine s, λ) + SVD	289	161
Malis and Vargas (C)	232	131
Malis and Vargas + SVD	294	189
Yang <i>et al.</i>	230	129
Yang <i>et al.</i> + SVD	291	187
SVD ($\tilde{\mathbf{H}}$)	810	149
SVD ($\tilde{\mathbf{H}}^T \tilde{\mathbf{H}}$)	722	126
Malis and Vargas (O)	1900	233

**Fig. 5** Gaussian kernel smoothed histograms of the rotation and translation errors for 10^6 runs on noise-free data

cases. Since refining both the depths and the scale factor gives the best performance, in the next experiment, we omit other refinements.

Note that computing the SVD of $\tilde{\mathbf{H}}^T \tilde{\mathbf{H}}$ is slightly faster than computing the SVD of $\tilde{\mathbf{H}}$ directly. This speed improvement likely occurs because the Eigen library optimizes SVD computations for symmetric matrices. However, this approach is less numerically stable due to the matrix $\tilde{\mathbf{H}}^T \tilde{\mathbf{H}}$ having larger condition number. Hence, the proposed closed-form solution is combined with SVD ($\tilde{\mathbf{H}}$). For the closed-form methods of Malis and Vargas and Yang *et al.*, we also applied the threshold to determine when to switch between the closed-form and SVD-based methods. The results of this version are denoted as Malis and Vargas + SVD, and Yang *et al.* + SVD. The proposed method, when combined with the refinement and the SVD-based approach, is stable and free of failures. In contrast, the other closed-form solutions even when combined with SVD still exhibits few failures, while having running time similar to that of the proposed method. In Figure 5, we show the Gaussian kernel smoothed histograms of the rotation and translation errors for 10^6 runs on noise-free data. The three closed-form methods are combined with SVD-based decomposition, and we omit the Malis and Vargas (O) implementation since it is not practical. The SVD-based decomposition methods are clearly more stable

than closed-form methods, with the proposed method outperforming existing closed-form approaches.

5 Conclusion

In this paper, we study the classical homography decomposition problem. We discuss the state-of-the-art closed-form and SVD-based approaches and proposed a novel closed-form method for homography decomposition that is based on the generation of 3D point correspondences. The proposed method is conceptually simple, geometrically meaningful, easy to understand, and implement and can be seen as an alternative to the existing closed-form methods Malis and Vargas Villanueva (2007) and Yang *et al.* (2018). To design a novel robust and fast method as well as to eliminate limitations of existing implementations of state-of-the-art methods, we in detail analyze different steps involved in homography decomposition methods. This includes fixing the median singular value of the homography matrix by computing one root of a cubic equation, the chirality check based on $\det(\mathbf{H})$, the Gauss-Newton refinement, possible failure cases and a simple way to recover $\mathbf{E}$ from $\mathbf{H}$. By analyzing the failure cases of the closed-form methods, we identify that they are related to the discriminant of the cubic equation used

to calculate the singular values. Based on this observation, we design a homography decomposition algorithm that combines the proposed closed-form solution with the SVD-based method, resulting in an approach that is both fast and stable. The source code is available at <https://github.com/yaqding/Homography-decomposition>

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